Namit Rana

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Experience

Research Assistant Aug 2025 – Present
Under Prof. Raj Kumar Sharma, GGSIPU *New Delhi, India*

- Trained **8 CNN architectures** on **5,666 dermoscopic images** (BCN20000 dataset); 3-layer CNN achieved **83% accuracy and AUC 0.90** for binary MEL/BCC skin cancer classification.
- Quantified dropout trade-off across all depth variants: AUC dropped to **0.77–0.81** with regularisation vs. **0.88–0.90** without, showing dropout hurt discrimination on this dataset.
- Co-authored and presented paper at **ICIDSSD '26, Jamia Hamdard**; received **Best Paper Award** among all submissions.

Technology Research Intern Feb 2025 – Apr 2025
To-Let Globe *New Delhi, India*

- Benchmarked **10+ LLMs** (GPT, Gemini, LLaMA, DeepSeek) on cost, latency, and fine-tunability to identify the optimal chatbot solution for internal deployment.
- Ran ML experiments on proprietary company datasets, evaluating open-source vs. proprietary model trade-offs; findings directly informed the final LLM selection in the recommendation report.
- Delivered a structured report detailing LLM selection, deployment strategy, and cost-performance trade-offs; served as the technical basis for the company's chatbot implementation.

Technology Intern – ML Recommendations Team Jun 2024 – Dec 2024
Sam Monitoring Solutions *New Delhi, India*

- Benchmarked ML models (Decision Trees, CNNs) on client datasets to generate operational recommendations; identified and fixed a **critical data leakage bug**, improving pipeline reliability.
- Proposed optimization strategies that yielded a **25% efficiency improvement**, validated by senior engineers across clients.

Projects

Glassbox – Structured Reasoning via LLM Post-Training | *JAX, Tunix, GRPO, TPU* | Kaggle Writeup

- Built a **3-tier post-training pipeline** (SFT → GRPO Reinforcement Learning → Self-Distillation) on Gemma-2-2B using JAX/Tunix, achieving ∼**100% format compliance** and **emergent self-correction behavior** on holdout sets.
- Curated a **50k-sample multi-domain reasoning corpus** (10k each across math, science, coding, logic, long-context) with strict 3-tier data hygiene — using **non-overlapping subsets per tier** to prevent leakage and ensure robust generalization.
- Engineered **dynamic KV-cache bounding** and XLA-static bucketing to fit entire SFT+RL+Distillation pipeline within a **single Kaggle TPU session** (9hr limit); replaced PPO with **GRPO to eliminate critic model**, removing OOM errors under bfloat16 + LoRA constraints.

Cardi-HACK - Ensemble ML Pipeline for Cardiogenetic Risk Prediction | *Python, Scikit-learn, Ensemble Learning*

- Achieved **Top 1% global ranking** using a stacked ensemble with probability calibration, handling extreme class imbalance in hypertrophic cardiomyopathy risk prediction.
- Trained models on **11 clinical features and ~300 SNPs**, applying MAF-based SNP filtering and robust feature normalization to handle noisy, high-dimensional medical data.

Leadership and Achievements

Honorable Mention - *Google × Kaggle Tunix Hackathon* (out of 11,173 entrants, 322 final submissions) Apr 2026
Research Paper Author & Presenter - "A Comparative Study of CNN Depth and Dropout Effects on Melanoma and Basal Cell Carcinoma Detection" - *ICIDSSD '26, Jamia Hamdard* - **Best Paper Award** Apr 2026
1st Place Winner (**Top 0.25%** among 2,000+ teams) - *Vihaan 8.0 Hackathon*, IEEE DTU Apr 2025
2nd Runner-Up (**Top 0.5%** among 600+ teams) - *HackWith Gujarat*, State-Level Hackathon Jun 2025
Finalist - Multiple National Hackathons (Govt. of Delhi, IIIT-D, GGSIPU, BVCOE, etc) 2024–2025
Hacktoberfest Super Contributor – Submitted **8 PRs**; **7 accepted and merged** across ML and Python repos Oct 2025

Education

University of Delhi Aug 2023 – May 2026
B.Sc. (Hons) Computer Science *New Delhi, India*

Technical Skills

Languages: Python, C++, SQL, JavaScript
Machine Learning: PyTorch, JAX, CNNs, Transformers, LLM Fine-Tuning, LoRA, PEFT, Gemma, GRPO, PPO
Data & Analytics: NumPy, Pandas, Scikit-learn, XGBoost, LightGBM, Feature Engineering
ML Systems & Infrastructure: GPU/TPU Training, Model Serving, Experiment Tracking (W&B/MLflow), Model Evaluation
Full-Stack & Deployment: MERN Stack (MongoDB, Express, React, Node), REST APIs, Docker, JWT Auth